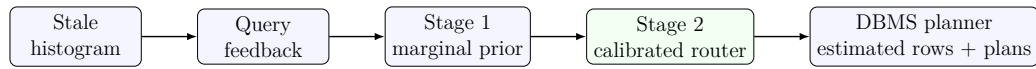


Graphical Abstract

OASIS: Regime-Aware Feedback Calibration for Optimizer Marginal Statistics

Author Names Redacted for Review



Highlights

OASIS: Regime-Aware Feedback Calibration for Optimizer Marginal Statistics

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- OASIS separates marginal prior generation from optimizer-facing feedback calibration.
- Stage-2 calibration is regime-dependent: hard projection is safer for joins/-planners, while consistency-gated soft projection improves future-predicate accuracy.
- A residual-gated calibrated router improves single-column and optimizer-facing accuracy with no join/planner regression relative to the deployed Hybrid.
- The calibration layer is estimator-agnostic across stale, heuristic, STHoles, QuickSel-H, and OASIS-MLP priors.
- Deployment safety is evaluated with traces, feedback noise, and PostgreSQL planner checks.

OASIS: Regime-Aware Feedback Calibration for Optimizer Marginal Statistics

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Abstract

Cost-based optimizers consume marginal statistics, but feedback-driven repairs of stale histograms are not automatically safe to expose to a planner. A learned or heuristic repair may improve future single-column selectivity while remaining inconsistent with the predicates just observed; downstream composition, join estimators, and DBMS plan selection can amplify that local inconsistency. This paper presents OASIS, a statistics-layer framework that separates marginal prior generation from optimizer-facing feedback calibration. A Stage-1 repair supplies a post-drift single-column prior, and a Stage-2 calibration layer turns that prior into a feedback-consistent statistics object before it is consumed downstream. OASIS maps engine-native statistics to a canonical quantile/CDF view, requires no base-table rescans, and leaves the optimizer’s search procedure and cost model unchanged.

The central contribution is the Stage-2 calibration layer. Hard ISOMER/IPF-style projection is safe for joins and planner-facing use, where residual one-dimensional errors are amplified, but it can overfit the current feedback window on future single-column predicates. We therefore add a consistency-gated soft projection that down-weights feedback contradicted by the most recent observations, and a residual-gated calibrated router that selects per case among stale, local-feedback, repaired, hard-projected, and soft-projected marginals using only deployment-visible feedback residuals. A banded relaxation of the hard constraints is a negative result, showing that the gain comes from regime-aware calibration and routing rather than from simply loosening equality constraints.

The evaluation is optimizer-facing rather than runtime-facing. First, calibration is necessary for safety: the raw learned marginal gives weak and unstable multi-column gains and is *actively harmful* in the bilinear Factor-Join join kernel, whereas calibrated OASIS improves every member of a six-

estimator composition family by 22%–38%. Across 12 PostgreSQL planner-only configurations, calibrated OASIS reduces aggregate row-estimation Q-error from 27.748 to 2.360 and matches the fresh-statistics plan shape on 96.1% of queries, while calibration reduces new plan deviations from 13.8% to 1.1%. Second, calibration is regime-dependent: the calibrated router lowers future-predicate single-column Q-error from 1.442 (hard projection) to 1.395 and optimizer-proxy selectivity Q-error from 1.446 to 1.396, while matching the safe hard/Hybrid behavior on FactorJoin and PostgreSQL checks because the gate does not route to soft where exactness matters. Third, calibration is estimator-agnostic: a Stage-1 swap experiment shows that the same Stage-2 router improves stale, LinInterp, FeedAvg, STHoles, QuickSel-H, and OASIS-MLP priors. The learned Stage 1 remains useful as a strong prior generator—under fixed projection it improves held-out future-predicate Q-error from 1.536 (projection from stale) to 1.446—but the deployment-facing novelty is the reusable calibration layer. No query runtime is measured.

Keywords: Statistics correction, Cardinality estimation, Histogram calibration, Cost-based optimizer, Query feedback, Online learning

1. Introduction

Cost-based optimizers (CBOs) choose query plans from estimates of predicate selectivity, intermediate cardinality, and operator cost. In many analytical database systems, these estimates rely on column-level statistics such as histograms, most-common-value lists, and density summaries. The statistics are periodically refreshed through **ANALYZE**-style procedures that read a full or sampled view of the table. This maintenance strategy is simple and robust, but it leaves a persistent staleness gap: tables continue to receive inserts, deletes, and updates while the optimizer keeps planning with the last statistics snapshot. As the gap widens, stale histograms induce systematic selectivity errors that can propagate through the optimizer’s cost model.

Existing remedies operate at different layers of the optimization stack. Frequent **ANALYZE** reduces staleness but increases scan overhead. Learned cardinality estimators predict point cardinalities for particular queries and often require raw data, query workloads, or table-specific training. Plan-level feedback systems adjust plan choices or hints after observing execution behavior, but they do not repair the underlying statistics objects consumed by the optimizer. This paper focuses on a complementary layer: *statistics-*

level correction. Rather than replacing the optimizer or learning complete query cardinalities, we update the column statistics supplied to the existing CBO.

The key observation is that a DBMS already obtains useful feedback while executing ordinary queries. For a filter predicate, the optimizer’s estimated selectivity and the executor’s observed selectivity expose a local discrepancy between the stale histogram and the current data distribution. The tempting response is to repair the histogram directly from those observations. However, the deployment problem is sharper than producing a marginal that looks better on average: a repaired marginal must also be reliable near the predicates that were just observed, because downstream estimators and planners may amplify local one-dimensional mistakes into larger joint-cardinality or plan-shape errors.

This paper treats stale marginal-statistics correction as two coupled tasks: *prior generation* and *feedback calibration*. A prior generator extrapolates from the stale histogram and recent observations to a plausible post-drift marginal. In OASIS we instantiate this first task with a compact 38K-parameter pooled MLP trained once on simulation-derived drift scenarios and then reused across columns and schemas. The learned prior is useful: under a fixed projection operator, initializing from the learned marginal lowers held-out future-predicate Q-error from 1.536 (projection from stale, i.e., ISOMER) to 1.446, and the advantage grows with drift. But prior quality alone is not sufficient. The raw learned marginal can be globally plausible while still violating the current feedback window, and our downstream experiments show that this mismatch is enough to make the raw marginal weak for composition and actively harmful in a bilinear join kernel.

The main contribution of OASIS is therefore the second task: an optimizer-facing Stage-2 calibration layer. The layer starts from a marginal prior, whether learned or heuristic, and returns a feedback-calibrated marginal statistics object without changing the optimizer’s search procedure or cost model. Its hard operator is an ISOMER/IPF-style projection that exactly matches the active feedback constraints and is the safe choice for join and planner-facing use. Exact matching, however, is not uniformly best: on future single-column predicates it can overfit the feedback window. We therefore add a consistency-gated soft projection that regularizes toward the prior while down-weighting feedback contradicted by the most recent observations. Because no single operator dominates, the deployed Stage 2 is a residual-gated *calibrated router* that chooses among stale, local-feedback, re-

paired, hard-projected, and soft-projected marginals using only the current feedback-window residual. A banded relaxation of the equality constraints is a negative result: widening the bands monotonically worsens accuracy, isolating the gain to regime-aware calibration and routing rather than to looser matching.

This calibration layer is not a patch for one particular learned model. A Stage-1 swap experiment feeds stale statistics, simple heuristics, STHoles, QuickSel-H, and the OASIS MLP through the same Stage-2 protocol. The calibrated router improves every raw prior in aggregate and reduces feedback residuals for all of them, with final Q-error clustering near 1.39–1.44 across very different prior sources. This result changes the interpretation of the two-stage design: the learned Stage 1 is a useful and necessary prior source for OASIS, especially for drift extrapolation and out-of-distribution generalization, but the reusable systems contribution is the calibration layer that makes imperfect marginals safe to consume.

The evaluation follows this scope. We evaluate the mechanism where it acts: future selectivity estimates, structural histogram error, downstream marginal composition—including a six-estimator composition family and the FactorJoin join kernel that consume OASIS as a drop-in marginal source—and real PostgreSQL planner estimates and plan shapes. The PostgreSQL study is planner-only: true rows are obtained with `COUNT(*)`, and optimizer-facing evidence comes from `EXPLAIN (FORMAT JSON)` under injected statistics states, without measuring query runtime. To reduce dependence on a single drift simulator, we also evaluate out-of-distribution drift families and benchmark-inspired DML traces with explicit insert, update, and delete events.

The contributions are:

- We formulate feedback-driven marginal-statistics repair as a DBMS-facing calibration problem: given a stale histogram, a causal feedback window, and a marginal prior from any repair source, produce optimizer-consumable statistics without rescanning the table or changing the optimizer. We instantiate a PostgreSQL-style adapter that maps MCV+histogram statistics to a canonical quantile/CDF view and back.
- We design OASIS around a reusable Stage-2 calibration layer. The layer combines hard feedback-consistency projection, consistency-gated soft projection, and a residual-gated calibrated router that chooses per case from deployment-visible feedback residuals. A banded projection is reported as a negative result, separating routing/calibration gains from simple con-

straint relaxation.

- We instantiate Stage 1 with a lightweight learned marginal-prior generator and show why a useful prior still matters: under a fixed projection operator the learned prior improves future-predicate Q-error over projection from stale statistics, and the raw learned prior generalizes well on strong out-of-distribution batch drift. At the same time, a Stage-1 swap shows that Stage 2 is estimator-agnostic across stale, heuristic, STHoles, QuickSel-H, and OASIS-MLP priors.
- We show that calibration is necessary for downstream and planner safety. Raw repaired marginals are not feedback-consistent, give weak and unstable composition gains, and are actively worse than stale in the FactorJoin join kernel; calibrated marginals improve every estimator in a six-member composition family and reduce PostgreSQL new plan deviations from 13.8% to 1.1%.
- We give an optimizer-facing evaluation, without query runtime, spanning single-column drift, structural and initial-distribution generalization, a six-estimator composition family, the FactorJoin join kernel, a 12-configuration PostgreSQL planner-only statistics-injection study, and deployment-safety checks under sparse, noisy, and trace-grounded feedback.

2. Background and Problem Formulation

2.1. Column Statistics and the Staleness Gap

We model a column histogram as an equi-depth quantile summary with B buckets. The internal boundaries $v_1, \dots, v_{B-1}$ correspond to fixed quantile levels $p_1, \dots, p_{B-1}$. A range predicate such as $x < c$ is estimated by evaluating the piecewise-linear CDF implied by those boundaries. Although real DBMS statistics may combine histograms with auxiliary summaries such as most-common-value lists, density vectors, or null fractions, OASIS operates on a canonical one-dimensional CDF view.

Let H_0 denote the stale histogram captured at the last statistics refresh and H^* denote the current post-drift distribution. For a predicate σ , the optimizer derives an estimate $\hat{s}_{H_0}(\sigma)$ while the executor can later reveal an actual selectivity $s^*(\sigma)$. We measure selectivity error using Q-error:

$$\text{QErr}(\hat{s}, s^*) = \max \left(\frac{\hat{s}}{s^*}, \frac{s^*}{\hat{s}} \right). \quad (1)$$

2.2. Statistics Correction

Given a stale histogram H_0 and an observation window $\mathcal{O} = \{o_1, \dots, o_K\}$, where each observation records a predicate type, predicate boundary, estimated selectivity, and actual selectivity, the goal is to produce corrected statistics $\hat{H}$ such that future predicates are estimated more accurately:

$$\mathbb{E}_\sigma [\text{QErr}(\text{CDF}_{\hat{H}}(\sigma), s_\sigma^*)] < \mathbb{E}_\sigma [\text{QErr}(\text{CDF}_{H_0}(\sigma), s_\sigma^*)].$$

The expectation is over future predicates from the workload distribution.

The solution must satisfy three deployment constraints. First, it must avoid table rescans, deriving $\hat{H}$ only from H_0 and recent feedback. Second, it must be cheap enough to run near query-planning time. Third, it must degrade gracefully when feedback is sparse or uninformative, preserving the stale prior rather than applying large unsupported corrections.

2.3. Scope

OASIS targets single-column histogram staleness. It does not repair multi-column correlation errors, join-key dependence, physical design problems, or cost-model calibration. These terms remain in the optimizer’s error budget even if the single-column histogram is accurate. The role of OASIS is to improve the marginal statistics that many downstream estimators and optimizers already consume.

3. OASIS Design

3.1. System Overview

OASIS sits between the statistics store and the optimizer. When the optimizer requests a histogram, OASIS reconstructs a canonical quantile representation, retrieves a recent observation window for the column, obtains a marginal prior from a Stage-1 repair source, calibrates that prior with the Stage-2 router, and returns an engine-native statistics object to the CBO. Query execution remains unchanged except for lightweight per-conjunct feedback collection.

3.2. Feature Representation

The correction model consumes a fixed tensor containing four blocks. The prior block stores the normalized quantile boundaries of H_0 . The meta block stores the null fraction, observation count ratio, and bucket count. The

observation block stores up to K recent predicate observations; each observation contains a predicate-type encoding and numeric features for predicate bounds, estimated selectivity, actual selectivity, and recency. The mask block marks valid observation slots. In our implementation $B = 10$ and $K = 16$, producing a compact input representation.

3.3. Stage 1: Marginal Prior Generation

The Stage-1 component supplies a marginal prior to the calibration layer. In the main OASIS instantiation, this prior is produced by a pooled MLP with a prior encoder, an observation encoder, an attention-like pooling layer over the observation window, and a residual output head that predicts corrections to the internal quantile boundaries. Residual prediction encourages conservative behavior when the feedback window is uninformative. The predicted boundaries are clamped to the normalized domain and projected into monotone order to ensure a valid CDF.

The model is trained offline on simulation-derived compound drift. Each training sample contains a stale prior, a causal feedback window, and post-drift quantile targets. Once trained, the same checkpoint is reused across columns in the evaluation; no per-table or per-distribution retuning is used. The Stage-2 calibration layer is not tied to this particular prior source: Section 5.5 also feeds stale, heuristic, STHoles, and QuickSel-H marginals into the same calibration protocol.

3.4. Feedback Collection and Integration

OASIS requires two integration points. A query-completion listener records predicate-level feedback from executor row-count instrumentation. A statistics-assembly hook intercepts the histogram before it reaches the optimizer and substitutes the corrected version. Because the correction is inserted at the statistics layer, the optimizer’s search procedure and cost model remain unchanged.

3.5. Statistics-Format Conversion

Many DBMSs do not expose a standalone editable equi-depth histogram. PostgreSQL, for example, couples histogram boundaries with MCV lists and other per-column summaries. We instantiate the interface for PostgreSQL-style statistics and design it to isolate engine-specific adapters. The adapter has three phases: reconstruct a canonical distribution view from the native object, apply correction on the canonical view, and decompose the corrected

view back into native statistics fields. This paper validates the PostgreSQL-style MCV+histogram instantiation; other engines require their own adapter logic.

3.6. Stage 2: Regime-Aware Feedback Calibration

The Stage-2 component takes a marginal prior and calibrates it against the current feedback window before the marginal reaches a downstream estimator or a DBMS planner. When the prior is the OASIS MLP output, we call the raw stage-one-only marginal OASIS-noProj; when hard projection is applied to that prior, we call the result OASIS. When the same hard projection is initialized from the stale histogram, it corresponds to the local-feedback ISOMER baseline. This separation is important: prior generation controls how far the correction can extrapolate beyond observed predicates, while Stage 2 controls whether the statistics object is safe to compose, join, or inject into a planner. Stage 2 remains a single-column calibration layer; it does not estimate multi-column dependence.

Hard projection.. The default Stage-2 operator builds the interval partition induced by the active feedback predicates, initializes cell masses from the supplied marginal prior, and performs cyclic I-projections so that every active predicate’s interval mass matches its observed selectivity exactly. When sequential drift makes the active set inconsistent, the oldest constraints are dropped first, an implicit stale-feedback filter. We denote the resulting two-stage output OASIS; it is the safe form for downstream composition and the DBMS planner when the supplied prior is the learned OASIS marginal.

Consistency-gated soft projection.. Exact matching is not always optimal. On *future* single-column predicates, matching a handful of recent feedback intervals exactly can overfit them; a softer calibration that regularizes toward the learned marginal generalizes better. We therefore add a soft Stage-2 operator that minimizes $\text{KL}(p \parallel p_{\text{prior}}) + \lambda \sum_j w_j (A_j p - y_j)^2$ over the same partition, where A_j is a feedback interval mask and y_j its observed selectivity. The crucial component is the weight w_j : rather than discarding old feedback by age, we discard it by *inconsistency with the current data state*. We fit a reference distribution to the most recent observations, score each constraint by its residual against that reference, and set $w_j = \exp(-(\text{conflict}_j/\tau)^2)$, so feedback that the recent observations contradict is suppressed while old-but-still-consistent feedback is retained. This keeps the single-column general-

ization benefit of soft calibration without softly fitting stale, contradicted constraints.

A negative result: banded projection.. A natural alternative is to relax the equality constraints into confidence bands $y_j \pm \delta_j$ and I-project the learned prior onto that polytope, so that $\delta_j = 0$ recovers hard projection. This does not help: widening the band monotonically worsens single-column accuracy (Section 5.4). The soft operator’s advantage comes from a better-conditioned cell-level optimization—it attains *lower* feedback residual than hard projection on single-column data—not from relaxed matching, so relaxation is the wrong axis.

The deployed Stage 2: a calibrated router.. No single operator dominates: hard projection is best on the join kernel and the planner, soft projection is best on future single-column and optimizer-facing predicates. We resolve this with a residual-gated *calibrated router* that selects, per case, the candidate with the lowest feedback residual on the current observation window—a deployment-visible signal with no fresh labels—among the stale prior, ISOMER, OASIS-noProj, hard projection, and consistency-gated soft projection. Because the gate is a residual minimizer over a pool that contains hard projection, the router can only improve on the residual signal, and it reduces to hard projection/ISOMER on workloads (joins, the PostgreSQL planner) where soft projection has higher residual. The output of this router is the deployed OASIS; the simpler Hybrid over stale, ISOMER, OASIS-noProj, and OASIS remains available where the soft candidate is unnecessary.

4. Experimental Methodology

The evaluation asks six questions. First, does the statistics-format conversion preserve probability mass and selectivity estimates? Second, how much does a learned marginal prior improve single-column histogram quality under drift? Third, when is Stage-2 calibration needed for downstream composition and planner safety? Fourth, is the calibration operator regime-dependent, and can a deployment-visible router select among hard and soft variants safely? Fifth, is the Stage-2 calibration layer tied to the OASIS MLP or reusable across other marginal priors? Sixth, do corrected statistics affect a real PostgreSQL optimizer pipeline in the intended direction, and what safety checks are needed when feedback is sparse, noisy, or trace-like rather than drawn from the training generator?

Unless otherwise stated, OASIS is trained once on Gaussian compound drift with $q \in \{1, 3, 5, 10, 15, 20\}$ and evaluated on held-out drift intensities $q \in \{1, 3, 5, 10, 15, 20, 25, 30\}$. All feedback-driven methods consume the same observation budget $K = 16$ and emit corrected quantiles on the same $B = 10$ grid. We compare against the stale prior, STHoles [1], ISOMER [2], QuickSel-H [3], and simple feedback heuristics. Metrics include Q-error, selectivity MAE, quantile MAE, feedback residual, PostgreSQL estimated-row Q-error, and plan-shape match to the fresh-statistics planner. For deployment-sensitivity diagnostics that require many perturbed runs, we also use a transparent optimizer-decision proxy: future predicates are generated from held-out distributions, each statistics state selects scan and join choices under a fixed cost proxy, and the selected choice is scored by its true proxy cost. This proxy is used only to analyze feedback-budget and feedback-noise behavior; it is not a runtime benchmark.

5. Single-Column Statistics Correction

5.1. Format-Conversion Validation

The PostgreSQL-style MCV+histogram conversion preserves total probability mass exactly in the tested settings. Round-trip residual-quantile MAE is at machine precision, with mean 2.4×10^{-17} and maximum 9.2×10^{-17} . Selectivity estimates remain consistent before and after conversion, with global MAE 1.1×10^{-4} and worst-case error below 0.39%. With PostgreSQL-like defaults, the full round trip completes in 0.066–0.073 ms. This validates the PostgreSQL-style conversion as a low-overhead bridge between native statistics and the canonical OASIS view.

5.2. Q-Error Across Drift Intensity

Figure 1 reports the main single-column result. Because direct single-column evaluation has no downstream composition to violate, this section isolates the learned repair stage (OASIS-noProj); the full two-stage OASIS is evaluated where the projection matters, in Sections 6.3–6.6, and compared on single-column accuracy in Table 4. At mild drift ($q = 1$), ISOMER is strongest and OASIS-noProj remains close to the stale prior, consistent with the graceful-degradation objective. From $q \geq 5$, OASIS-noProj is the best method under the shared feedback budget. At $q = 10, 20$, and 30 , it reduces Q-error over the stale prior by 51.3%, 57.9%, and 60.6%, respectively, outperforming ISOMER in this moderate-to-large drift regime. ISOMER is

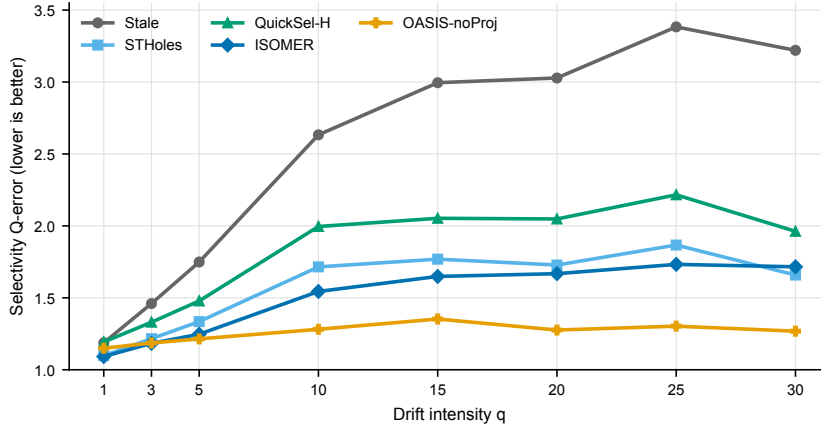


Figure 1: Single-column selectivity Q-error across drift intensity. All methods consume the same feedback budget. This figure isolates the learned repair stage (OASIS-noProj), which remains close to the stale prior at mild drift and becomes strongest as drift grows; the full two-stage OASIS is compared in Table 4.

constrained by the observed feedback window rather than a learned global drift prior. The advantage persists on unseen drift intensities $q = 25$ and $q = 30$.

Table 1 compares OASIS-noProj with simple feedback heuristics. Fixed linear interpolation and exponential feedback averaging are substantially weaker than learned correction; FeedAvg worsens estimation in all tested settings. This shows that the gains are not simply due to using feedback, but to learning how the prior and observations should interact.

5.3. Structural Metrics and Initial-Distribution Generalization

Table 2 shows that OASIS-noProj improves structural histogram quality, not only Q-error. At $q = 10$, quantile MAE drops from 0.208 for the stale prior to 0.055 for OASIS-noProj, and selectivity MAE drops from 0.190 to 0.049. OASIS-noProj remains the best method on both metrics for every $q \geq 5$.

Table 3 evaluates transfer across initial distributions at $q = 10$. The learned stage (OASIS-noProj), trained on Gaussian-mixture initial states, remains best across uniform, skewed, bimodal, triangular, exponential, and Gaussian-mixture starts, reducing Q-error by 44%–61% over the stale prior.

The single-column results so far isolate Stage 1. Table 4 adds the deployed two-stage OASIS. Applying the Stage-2 projection costs only a small

Table 1: Simple heuristic baselines vs. OASIS-noProj (the learned stage-1 repair) at selected drift intensities. Results are averaged over three independent training seeds (42, 123, 456) with a different evaluation protocol than the main single-column comparison; absolute Q-Error values are therefore not directly comparable across the two protocols. Both heuristics consume the same observation window $K=16$. FeedAvg worsens estimation at all drift levels.

q	Q-Error ($\downarrow$ better)			
	Stale	LinInterp	FeedAvg	OASIS-noProj
5	1.791	1.628	1.835	1.238
10	2.471	2.118	2.565	1.255
20	3.370	2.749	3.553	1.296
30	3.500	2.823	3.690	1.345

Table 2: Quantile MAE and Selectivity MAE across drift intensities from the clean synthetic rerun. Bold denotes best method per metric.

q	Quantile MAE ($\downarrow$)					Selectivity MAE ($\downarrow$)				
	Prior	STHoles	QSel-H	ISOMER	OASIS-noProj	Prior	STHoles	QSel-H	ISOMER	OASIS-noProj
1	0.043	0.027	0.036	0.023	0.041	0.038	0.024	0.031	0.020	0.035
3	0.106	0.064	0.071	0.047	0.048	0.095	0.057	0.064	0.040	0.042
5	0.145	0.085	0.094	0.057	0.054	0.131	0.077	0.085	0.051	0.048
10	0.208	0.115	0.126	0.080	0.055	0.190	0.105	0.114	0.072	0.049
15	0.230	0.119	0.128	0.094	0.062	0.203	0.104	0.113	0.083	0.054
20	0.237	0.115	0.127	0.102	0.052	0.214	0.104	0.115	0.091	0.046
25	0.248	0.116	0.131	0.095	0.053	0.223	0.104	0.118	0.084	0.046
30	0.231	0.103	0.111	0.103	0.049	0.204	0.091	0.098	0.090	0.042

amount of single-column accuracy relative to OASIS-noProj (aggregate Q-error 1.307 versus 1.254), because direct single-column evaluation has no downstream estimator to violate. At mild drift $q = 1$, ISOMER remains lower (1.092 versus 1.111), but from $q \geq 3$ the full OASIS stays below ISOMER. Thus the deployed form preserves a learned single-column advantage in the moderate-to-large drift regime; this small projection cost is the price paid for the downstream composition and join safety established in the following sections.

5.4. Stage-2 Calibration Variants and the Calibrated Router

The previous subsection used hard projection. We now study the Stage-2 calibration directly. This experiment reuses the cached drift data and the same OASIS checkpoint, and evaluates each Stage-2 variant on held-out

Table 3: Q-Error across different initial distributions at $q=10$. A model trained only on Gaussian-mixture initial states remains best on all tested distributions.

Initial Distribution	Stale	STHoles	QuickSel-H	ISOMER	OASIS-noProj
Gaussian Mixture	2.497	1.453	1.845	1.493	1.262
Uniform	2.409	1.503	1.977	1.534	1.284
Skewed (Power-law)	3.517	1.679	2.202	1.609	1.379
Bimodal	2.488	1.554	2.069	1.462	1.276
Triangular	2.225	1.420	1.833	1.505	1.238
Exponential	3.097	1.592	2.141	1.593	1.292

Table 4: Single-column selectivity Q-error ($\downarrow$) of the full two-stage OASIS versus the OASIS-noProj ablation, on in-distribution compound drift. At mild drift $q = 1$, ISOMER is lowest; from $q \geq 3$, the full OASIS stays below ISOMER, so the deployed (projected) form preserves a learned single-column advantage in the moderate-to-large drift regime. The projection costs only a small amount relative to the stage-1 ablation, the price paid for the downstream composition and join safety shown in Tables 9, 10 and 11.

q	Stale	ISOMER	OASIS-noProj	OASIS	OASIS red.%
1	1.188	1.092	1.150	1.111	+6%
3	1.460	1.184	1.186	1.178	+19%
5	1.749	1.247	1.215	1.209	+31%
10	2.633	1.544	1.281	1.393	+47%
15	2.995	1.649	1.353	1.409	+53%
20	3.028	1.668	1.276	1.364	+55%
25	3.383	1.733	1.303	1.412	+58%
30	3.220	1.716	1.268	1.379	+57%

future predicates per case (geometric-mean selectivity Q-error), together with the feedback residual on the observation window, quantile MAE against fresh statistics, and an optimizer-proxy join regret. It is a controlled comparison of calibration operators for a fixed learned prior, not a new training protocol.

Table 5 shows the regime structure. Full-window soft projection has the lowest single-column Q-error (1.376 versus hard 1.442) and the lowest feedback residual, confirming that softening helps single-column generalization rather than hurting it. But full-window soft is unsafe under sequential drift: it softly fits stale, contradicted feedback, which (in separate runs) inflates out-of-distribution batch-load Q-error to 1.482 versus hard 1.272. A fixed recent window ($k=8$) removes that failure but discards old-but-still consistent feedback, giving back accuracy (1.408). The consistency-gated soft variant keeps

Table 5: Stage-2 calibration variants on held-out future predicates (cached drift, fixed OASIS checkpoint). Lower is better. Full-window soft has the lowest selectivity Q-error and feedback residual but is unsafe under sequential drift (Section 5.6); the calibrated router is the best *deployable* variant, recovering most of the single-column gain over hard projection while remaining safe to route.

Stage-2 variant	Sel. QErr	Feedback resid.	Quantile MAE	Join regret
Hard projection (OASIS-Proj)	1.442	0.0440	0.0519	1.0344
Full-window soft	1.376	0.0348	0.0483	1.0284
Recent-window soft ($k=8$)	1.408	0.0404	0.0471	1.0322
Conflict-aware soft	1.401	0.0392	0.0473	1.0313
Hybrid (no soft)	1.429	0.0395	0.0513	1.0326
Calibrated router	1.395	0.0358	0.0486	1.0302

old feedback that the recent observations do not contradict and reaches 1.401 at a lower feedback residual than the fixed window. The calibrated router, selecting per case by feedback residual over {stale, ISOMER, OASIS-noProj, hard, conflict-aware soft}, is best overall at 1.395 among deployable variants, with low feedback residual (0.0358), the best deployable join regret (1.0302), and the lowest deployable new-risk loss (1.37% versus hard 1.81%); it routes to soft on 44% of cases and to hard/ISOMER/OASIS-noProj otherwise. As a negative control, a banded projection that relaxes the equality constraints into confidence bands $y_j \pm \kappa \sqrt{y_j(1 - y_j)}$ monotonically worsens single-column Q-error as κ grows (1.437, 1.450, 1.470, 1.509 for $\kappa \in \{0.04, 0.10, 0.20, 0.35\}$), so the gain comes from per-case routing, not from relaxed matching.

The same ordering holds on the optimizer-decision proxy of Section 4: selectivity Q-error / join regret are 1.446/1.0348 for hard projection, 1.404/1.0312 for conflict-aware soft, 1.430/1.0326 for the stale/ISOMER/noProj/hard Hybrid, and 1.396/1.0297 for the calibrated router—the best non-fresh result. Crucially, the router does not trade away safety: where soft projection has higher feedback residual it is never selected. On the FactorJoin kernel the residual-gated router stays at hard level (join Q-error 1.0174 versus hard 1.024) while soft projection alone is worse (1.0367); and on the PostgreSQL plan-shape batch (Section 6.6) the router is identical to the Hybrid. The calibrated router is therefore $\geq$ hard projection on single-column and optimizer-facing accuracy and equal to the deployed Hybrid on join and planner safety.

Table 6: Stage-1 prior-source swap under a fixed Stage-2 calibration protocol. Each row supplies a different single-column marginal prior to Stage 2; lower Q-error and feedback residual are better.

Stage-1 source	Raw QE	+Hard QE	+Router QE	Router gain	Raw resid.	Router resid.
Stale prior	2.404	1.539	1.437	40.2%	0.1206	0.0383
LinInterp	2.116	1.508	1.417	33.0%	0.1072	0.0379
FeedAvg	2.474	1.544	1.437	41.9%	0.1256	0.0383
STHoles	1.572	1.436	1.394	11.3%	0.0542	0.0365
QuickSel-H	1.614	1.423	1.392	13.7%	0.0573	0.0347
OASIS MLP	1.643	1.439	1.392	15.3%	0.0643	0.0358

5.5. Estimator-Agnostic Calibration

The preceding calibration study fixes the Stage-1 prior to the OASIS MLP. To test whether the Stage-2 layer is a reusable calibration mechanism rather than an OASIS-specific patch, we run a Stage-1 prior-source swap. The experiment uses the same cached drift cases, the same feedback windows, and the same future predicate protocol as Section 5.4. We vary only the marginal prior supplied to Stage 2: stale statistics, two simple feedback heuristics (LinInterp and FeedAvg), STHoles, QuickSel-H, and the OASIS MLP. For each prior we report the raw marginal, hard projection initialized from that prior, and the calibrated router over {stale, ISOMER, raw prior, hard projection, conflict-aware soft projection}.

Table 6 shows that Stage 2 is not tied to the learned OASIS prior. Hard projection improves every raw prior, and the calibrated router further improves every prior in aggregate, reducing future-predicate Q-error by 11.3%–41.9% relative to the corresponding raw marginal. Feedback residuals follow the same pattern: raw marginals have residuals between 0.0542 and 0.1256, while the routed marginals fall to 0.0347–0.0383. The router’s choices are also distributed across candidates rather than always selecting the OASIS MLP path: depending on the prior source, it routes to soft projection on 44%–61% of cases, to ISOMER on 19%–38%, to hard projection on up to 22%, and occasionally to the raw or stale candidate. Per-drift breakdowns show three small QuickSel-H cases where hard projection is slightly lower than the router, so the conclusion is aggregate/source-level rather than pointwise dominance over hard. The broader conclusion is nevertheless direct: Stage 2 is an estimator-agnostic calibration layer. The OASIS MLP remains useful as a learned prior generator, but the reusable optimizer-facing mechanism is the calibration layer that can accept multiple imperfect marginal sources.

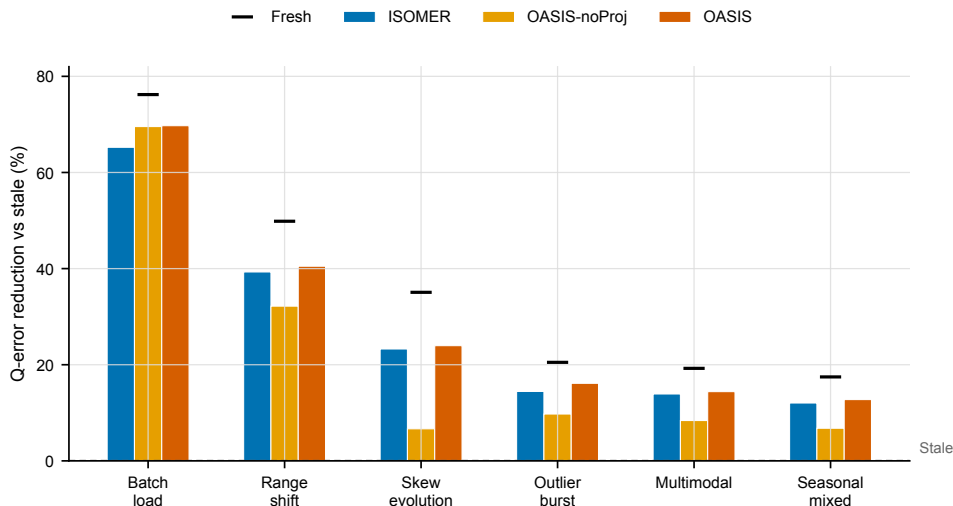


Figure 2: Out-of-distribution drift realism suite. OASIS is trained only on compound drift and evaluated unchanged on deployment-inspired drift families. Bars report the percentage reduction in geometric-mean selectivity Q-error relative to stale statistics for the same drift family; higher is better. The dashed zero line is stale, and black ticks show the fresh post-drift reference.

5.6. Out-of-Distribution Drift Realism

The main OASIS checkpoint is trained on compound drift. To test whether the learned correction is useful beyond that generator, we evaluate it unchanged on six deployment-inspired drift families: batch loads, range shifts, skew evolution, outlier bursts, multimodal drift, and seasonal/mixed drift. Each case starts from a stale histogram, interleaves the new drift family with 16 feedback observations, and evaluates future selectivity against the final post-drift distribution. No OOD retraining is used.

Figure 2 shows that the correction path remains useful under these unseen drift mechanisms. The strongest OOD stress case is batch loading, where stale Q-error is 4.203; OASIS-noProj reduces it to 1.280, outperforming ISOMER’s 1.461, and the full OASIS further improves to 1.272. On range shift, skew evolution, outlier, multimodal, and seasonal drift, OASIS is the best non-fresh method, with Q-error between 1.057 and 1.187. This experiment is not a substitute for production trace validation, but it shows that the gains are not confined to the compound-drift generator used for training.

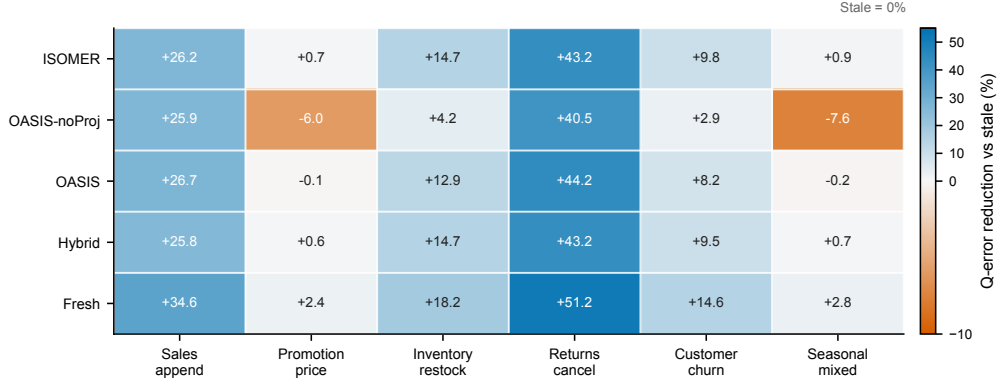


Figure 3: Benchmark-inspired DML trace sanity check. Each trace emits explicit insert/update/delete events anchored to an analytical benchmark maintenance pattern. Cells report the percentage reduction in geometric-mean selectivity Q-error relative to stale statistics for the same trace; higher is better, while negative values indicate regressions below stale. Stale is 0% by definition, and the Fresh row shows the post-drift reference.

5.7. Benchmark-Inspired DML Trace Sanity Check

The OOD suite changes the drift family, but it still uses synthetic drift operators. We therefore add a second realism check with explicit DML event streams anchored to analytical benchmark table/column patterns: append-heavy sales dates, promotion price revisions, inventory restocking, returns and cancellations, customer segment churn, and seasonal mixed maintenance. Each trace emits insert, update, and delete counts before a feedback observation is collected. These traces are not production logs or DBMS telemetry; they are a transparent sanity check that the correction layer remains useful when drift is produced by event sequences rather than by the compound training generator.

Figure 3 reinforces the same deployment pattern under event-sequence drift. Across the six traces, stale statistics have aggregate Q-error 1.295; the full OASIS reduces this to 1.076 and residual-gated Hybrid to 1.072. OASIS is the best non-fresh method on sales append and returns/cancellation traces, where the stale histogram is visibly shifted. On the milder price, inventory, customer, and seasonal traces, ISOMER is best or tied because enforcing the current feedback window is sufficient. OASIS-noProj can regress on these mild update-heavy traces, which supports the two-stage design: the learned stage is useful, but the statistics object passed to a planner should be the projected (full) OASIS or residual-gated Hybrid.

Table 7: Counterfactual selectivity evaluation on compound drift test data. “Sampled-Refresh” is a simulated column-level resampling proxy. Recovery measures the fraction of fresh/oracle Q-error improvement recovered by the learned stage (OASIS-noProj). No query runtime is measured.

q	#Pred	Stale QE	Sampled-Refresh	OASIS-noProj QE	Recovery
5	30	1.714	1.492	1.263	69%
10	30	2.720	2.121	1.232	77%
15	30	2.798	2.183	1.284	74%
20	30	2.779	2.201	1.389	64%
Mean	120	2.503	1.999	1.292	71%

6. Optimizer-Facing Evaluation

6.1. Counterfactual Selectivity Evaluation

We first isolate the causal effect of the statistics state on selectivity estimation. The same held-out drift samples are evaluated under stale, OASIS-noProj-corrected, and simulated sampled-refresh statistics, while the underlying post-drift distribution and predicates are held fixed. This is not a runtime benchmark; it measures selectivity Q-error under controlled statistics states.

Table 7 shows that OASIS-noProj recovers 64%–77% of the fresh/oracle improvement and substantially outperforms the sampled-refresh proxy. The result should be interpreted as per-predicate selectivity evidence, not as evidence of wall-clock query acceleration.

6.2. Does the Learned Stage Help After Projection?

A reasonable concern about the two-stage design is whether the learned stage contributes anything once the projection is applied: in several aggregate downstream tables the full OASIS and ISOMER are close. We isolate the question with a projection-initialization ablation. We hold the projection operator and the feedback constraints fixed and vary only the marginal that initializes the projection: initializing from the stale histogram reproduces ISOMER, initializing from the learned stage gives the full OASIS, and initializing from the post-drift marginal is a reference upper bound. All variants are scored on held-out future predicates, not on the feedback window they were fit to.

Table 8 shows that the learned stage is not decorative. Holding the projection fixed, initializing from the learned marginal lowers future-predicate

Table 8: Projection-initialization ablation on held-out future predicates (in-distribution compound drift; geometric-mean selectivity Q-error). The projection operator and feedback constraints are identical across columns; only the marginal that initializes the projection changes. **Proj(stale)** initializes from the stale histogram and is exactly ISOMER; **Proj(OASIS)** initializes from the learned stage and is the full two-stage OASIS; **Proj(fresh)** initializes from the post-drift marginal as a reference upper bound. Win frac is the fraction of held-out predicates on which the full OASIS beats ISOMER.

q	Stale	Proj(stale)	OASIS-noProj	Proj(OASIS)	Proj(fresh)	Fresh	Proj(OASIS) vs ISOMER	Win frac
1	1.224	1.137	1.271	1.181	1.104	1.000	-3.9%	43.3%
3	1.679	1.298	1.370	1.283	1.146	1.000	+1.2%	53.5%
5	2.004	1.371	1.511	1.348	1.160	1.000	+1.7%	55.0%
10	2.774	1.568	1.647	1.461	1.169	1.000	+6.8%	59.1%
15	3.103	1.735	1.894	1.616	1.167	1.000	+6.8%	56.6%
20	3.124	1.784	1.827	1.590	1.165	1.000	+10.9%	60.9%
25	3.380	1.763	1.911	1.601	1.173	1.000	+9.2%	58.3%
30	3.091	1.790	1.869	1.556	1.174	1.000	+13.0%	60.3%
All	2.415	1.536	1.645	1.446	1.157	1.000	+5.9%	55.9%

Q-error from 1.536 (ISOMER) to 1.446 (full OASIS), a 5.9% reduction, and the advantage grows monotonically with drift: +6.8%, +10.9%, and +13.0% at $q = 10$, 20, and 30, while the two methods are within noise at $q = 1$. Per-predicate, the full OASIS beats ISOMER on 55.9% of held-out predicates, up to 60.9% at $q = 20$. A threshold-only predicate population gives the same pattern more mildly (+4.3% overall, +8.8% at $q = 30$).

Two further readings support the claim that both stages are necessary. First, the columns form a clean initialization-quality ladder—stale 1.536, learned 1.446, fresh 1.157—so a better starting marginal yields a better projected result, and the learned initialization closes about a quarter of the ISOMER-to-fresh headroom. Second, neither stage alone reaches the two-stage result: the raw learned marginal (OASIS-noProj, 1.645) is worse than ISOMER on this metric, and ISOMER (projection from a stale start) is worse than the full OASIS. This also explains why aggregate downstream tables show ISOMER and the full OASIS closer: the learned stage’s measurable advantage is on the marginal’s future-predicate accuracy, and how much of it survives depends on how strongly a given downstream estimator’s error is dominated by the marginal. A finer locality decomposition is given in Appendix Appendix A.

6.3. Marginal-to-Joint Composition

This experiment studies composition rather than replacing multi-column cardinality-estimation methods. OASIS repairs one-dimensional marginals,

Table 9: Marginal-to-joint composition: average Q-error ($\downarrow$) for two-column range predicates. Values average configuration-level geometric means over 20 correlation/drift settings. OASIS is the full two-stage system; OASIS-noProj is the learned stage without the feedback-consistency projection; Hybrid uses only feedback-window residuals for selection.

Estimator	Stale	OASIS-noProj	ISOMER	OASIS	Hybrid	Fresh	OASIS-noProj +%	OASIS +%	Hybrid +%
Independence / max-entropy	1.820	1.679	1.220	1.222	1.214	1.185	+7.9%	+31.8%	+32.2%
Gaussian copula	1.558	1.513	1.185	1.176	1.166	1.139	+2.4%	+23.1%	+23.8%

so the question is how those repaired marginals behave when a downstream estimator combines them under a fixed dependence assumption. We generate correlated two-column tables with drift, correct each single-column marginal independently, and feed the marginals into two joint estimators: independence, the maximum-entropy estimate under only one-dimensional marginal constraints, and a Gaussian copula with fixed correlation structure. Fresh uses post-drift marginal inputs under the same downstream estimator; it is not an oracle joint distribution.

Table 9 gives the main diagnostic result. Despite the strong single-column performance of the learned stage, OASIS-noProj provides only modest and unstable gains after two-column composition: it improves average joint Q-error by 7.9% under independence and by 2.4% under the Gaussian copula. ISOMER is substantially stronger in this setting because it enforces current feedback constraints. The attribution explains the gap: OASIS-noProj improves marginal Q-error over stale in only 43.6% of composition rows and beats ISOMER on marginal Q-error in only 3.9%. Thus, the learned stage should not be interpreted as a stable multi-column correction method on its own.

The two-stage design resolves this directly: the projection stage calibrates the learned marginal before composition, without changing the dependence model. By projecting the OASIS-noProj marginal onto the feedback constraints, the full OASIS reaches Q-error 1.222 under independence and 1.176 under the Gaussian copula, nearly matching ISOMER and approaching fresh marginal inputs. Hybrid provides a conservative deployment policy by choosing among stale, ISOMER, OASIS-noProj, and OASIS using only feedback-window residuals.

6.4. Composition-Family Generalization

The two-column study above uses two downstream estimators. A natural concern is whether the deployment pattern—OASIS-noProj is an unsta-

Table 10: OASIS as a drop-in marginal upgrade across a family of composition estimators. Average joint Q-error ($\downarrow$) for two-column range predicates; dependence structure is held fixed per estimator and only the marginal input varies. The full two-stage OASIS and Hybrid improve every estimator, whereas OASIS-noProj (the learned stage without the feedback-consistency projection) is weak and unstable, confirming the projection is required across composition methods.

Estimator	Stale	OASIS-noProj	ISOMER	OASIS	Hybrid	Fresh	OASIS-noProj +%	OASIS +%	Hybrid +%
Independence	1.924	1.798	1.191	1.191	1.185	1.138	+6.6%	+38.1%	+38.4%
IPF/Sinkhorn-2D	1.668	1.564	1.302	1.280	1.274	1.248	+6.3%	+23.3%	+23.6%
Gaussian copula	1.645	1.558	1.251	1.224	1.216	1.182	+5.3%	+25.6%	+26.1%
Clayton copula	1.836	1.571	1.310	1.256	1.251	1.222	+14.4%	+31.6%	+31.9%
Gumbel copula	1.639	1.535	1.254	1.234	1.226	1.192	+6.4%	+24.7%	+25.2%
Frank copula	1.621	1.552	1.291	1.260	1.253	1.221	+4.2%	+22.2%	+22.7%

ble composition input, and the feedback-consistency projection is required—is specific to those two estimators or general across composition methods that consume single-column marginals. We therefore embed OASIS as a drop-in marginal source in a family of six independently designed composition estimators and vary only the marginal input. The family covers the independence/maximum-entropy estimator, iterative proportional fitting (IPF/Sinkhorn) on a two-dimensional grid, and four copulas with distinct dependence structure: Gaussian, Clayton, Gumbel, and Frank. To keep the comparison a clean marginal ablation and to respect the scope boundary that OASIS does not infer dependence, the dependence structure is held fixed and identical for every marginal input: each copula uses one association parameter derived from the data-generating correlation, and IPF seeds its cell-level association from a stale two-dimensional sketch and only reconciles it with each method’s corrected one-dimensional marginals. The Archimedean copulas are intentionally mis-specified models for the Gaussian-correlated data, which tests whether the marginal-repair benefit survives an imperfect dependence model.

Table 10 confirms both halves of the deployment story across the whole family. OASIS-noProj again gives only weak and inconsistent composition gains (between +4.2% and +14.4% over stale), whereas the full OASIS improves every estimator by 22.2%–38.1% and Hybrid by 22.7%–38.4%, in each case approaching the fresh-marginal input. The gain therefore does not depend on any single dependence model, and it persists even under the mis-specified Archimedean copulas. This generalizes the calibration finding from two estimators to a method family: the learned stage is a useful input to existing composition estimators, but the feedback-consistency projection is what makes it a stable one.

Table 11: OASIS embedded in the FactorJoin bin-based join kernel. Join-cardinality Q-error ($\downarrow$) for a two-table equi-join whose join-key distributions drift; only the single-column key histogram consumed by FactorJoin varies across methods. The full two-stage OASIS and Hybrid recover most of the stale-to-fresh join error, whereas OASIS-noProj (the learned stage without projection) is actively harmful—it exceeds the stale baseline in this bilinear kernel.

q	Stale	OASIS-noProj	ISOMER	OASIS	Hybrid	Fresh	OASIS +%	Hybrid +%
5	1.323	1.248	1.009	1.010	1.008	1.005	+23.6%	+23.8%
10	1.179	1.435	1.013	1.031	1.021	1.005	+12.6%	+13.4%
20	1.179	1.447	1.017	1.021	1.021	1.005	+13.4%	+13.4%
30	1.178	1.389	1.026	1.034	1.020	1.005	+12.3%	+13.4%
All	1.213	1.377	1.016	1.024	1.017	1.005	+15.6%	+16.1%

6.5. FactorJoin Integration

The previous composition studies remain single-table. To test whether corrected marginals propagate into a learned *join* cardinality estimator, we embed OASIS in the bin-based join kernel of FactorJoin [4]. For an equi-join $A.k = B.k$, FactorJoin partitions the join-key domain into bins and estimates the join size as $\sum_{\text{bin}} \text{cnt}_A[\text{bin}] \text{cnt}_B[\text{bin}] / \text{dv}[\text{bin}]$, where each table’s per-bin key frequency $\text{cnt}_T[\text{bin}]$ is read from a single-column histogram of its join-key distribution. We reimplement this kernel and make the per-table key histogram the OASIS-correctable object: when the key distribution drifts, a stale histogram yields wrong per-bin frequencies and a wrong join estimate. The two tables’ keys are generated independently, so the only quantity that varies across methods is the quality of each single-column key histogram; the binning, the distinct-values-per-bin assumption, and the row counts are identical for all methods. The metric is join-cardinality Q-error against the exact join size on the drifted data; no query runtime is measured.

Table 11 shows the strongest form of the calibration finding, and Figure 4 visualizes the deployable calibrated methods on a stale-to-fresh-normalized scale. In Table 11, OASIS-noProj does not merely underperform—it is actively harmful, raising aggregate join Q-error above the stale baseline (−13.5%, and as much as −22.7% at moderate drift). The mechanism is structural: the FactorJoin estimate is a bilinear functional of the two per-bin mass vectors, so it is sensitive to mass concentration rather than to coarse range selectivity. The learned stage, trained to minimize range-selectivity error, can return a directionally correct but over-concentrated histogram—occasionally with collapsed boundaries after monotone clamping—whose fine-

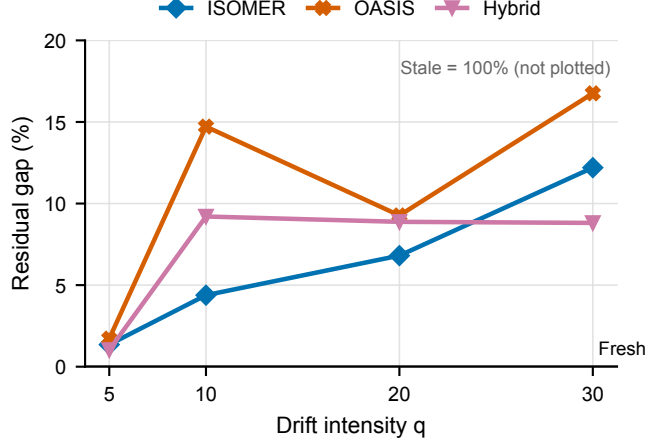


Figure 4: FactorJoin join-cardinality Q-error by drift intensity for the deployed/calibrated marginal sources. Values are the residual gap to the fresh-marginal floor, normalized by the stale-to-fresh gap; 0% is fresh and 100% is stale, so lower is better. The no-projection ablation is reported in Table 11; the full OASIS and Hybrid stay near the fresh-marginal floor.

grained mass profile inflates $\sum_{\text{bin}} p_A p_B$ and therefore the join estimate. The feedback-consistency projection mitigates this failure: the full OASIS improves over stale at every tested drift level (12.3%–23.6%, with 15.6% aggregate gain), Hybrid improves by 13.4%–23.8%, and both closely track the fresh-marginal input. The effect is stable as the join-bin count varies from 10 to 100. The broader lesson sharpens the deployment boundary: the more non-linear the downstream composition, the more essential feedback-consistency calibration becomes, and a join estimator should consume the full OASIS rather than the raw learned marginal.

6.6. PostgreSQL Planner-Only Statistics Injection

The previous experiments evaluate selectivity and downstream estimator behavior. We next validate that corrected statistics can affect a real DBMS optimizer pipeline. The PostgreSQL experiment creates a synthetic table with fresh current data, builds alternative single-column statistics for `fact.x`, and injects them into `pg_statistic` by copying typed statistics rows from analyzed source tables. For each query, true cardinality is obtained with `COUNT(*)`; PostgreSQL estimated rows and plan shape are obtained with `EXPLAIN (FORMAT JSON)`. The experiment does not measure query runtime.

Table 12: Multi-configuration PostgreSQL planner-only statistics-injection evidence. Values are means over configurations; parentheses show one standard deviation across configurations. True rows use `COUNT(*)`; estimated rows and plan shapes use `EXPLAIN (FORMAT JSON)`. No query runtime is measured.

Method	Row QE	Fresh Plan	Recovery	New Deviations
Stale	28.436 (6.585)	56.0% (5.5)	0.0% (0.0)	0.0% (0.0)
ISOMER	2.372 (0.283)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
OASIS-noProj	3.313 (0.228)	88.6% (2.7)	91.4% (3.2)	13.7% (2.0)
OASIS	2.375 (0.275)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
Hybrid	2.374 (0.286)	96.1% (1.9)	92.2% (3.9)	1.0% (1.0)
Fresh	1.415 (0.095)	100.0% (0.0)	100.0% (0.0)	0.0% (0.0)

Table 12 provides the strongest DBMS-facing evidence in this paper. The batch covers two drift directions, two table sizes, and three random seeds, yielding 12 PostgreSQL configurations. Across all query instances, stale statistics produce aggregate row Q-error 27.748 and match the fresh-statistics plan shape on only 56.0% of queries. OASIS-noProj reduces row Q-error to 3.306 and recovers 91.7% of stale/fresh plan-shape disagreements, demonstrating that learned marginal correction propagates into PostgreSQL’s native planner. However, the learned stage alone also introduces new plan deviations in 13.8% of cases where stale statistics already matched fresh.

The full OASIS largely removes this safety issue. It reduces aggregate row Q-error to 2.360, matches the fresh-statistics plan shape on 96.1% of queries, recovers 92.6% of stale/fresh plan-shape disagreements, and reduces new plan deviations to 1.1%. Across configurations, mean row Q-error is 2.375 ± 0.275 for OASIS, 2.372 ± 0.283 for ISOMER, and 3.313 ± 0.228 for OASIS-noProj. The planner-only result supports the two-stage design: the learned stage provides the global marginal correction, while the projection stage calibrates it before the DBMS planner consumes it.

Figure 5 breaks the PostgreSQL result down by query family. Stale statistics disagree with the fresh-statistics plan on 95/336 selection queries, 215/336 join queries, and 134/336 join-with-dimension-filter queries. The full OASIS matches the fresh plan on 98.5% of selections, 92.3% of joins, and 97.6% of filtered joins. The safety benefit over the learned stage alone is clearest in new deviations: OASIS-noProj creates new plan-shape deviations in 7.5%, 19.8%, and 17.8% of the three families, whereas the full OASIS reduces these rates to 0.0%, 5.0%, and 0.0%.

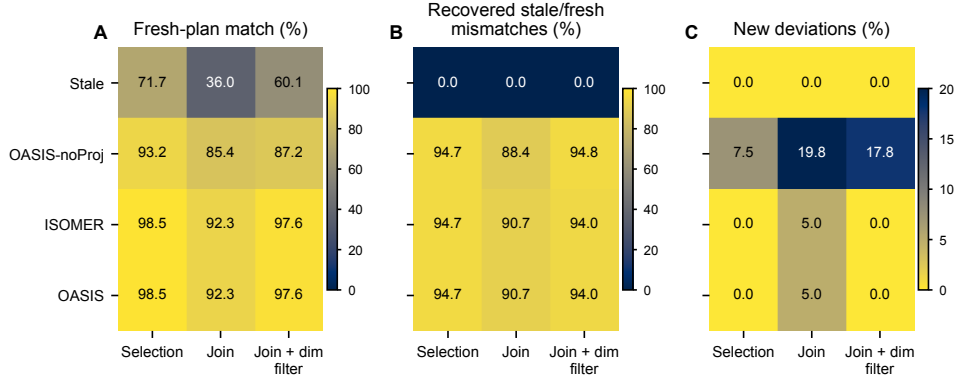


Figure 5: PostgreSQL planner-only plan-shape effects by query family. Cell labels are percentages. Panel A reports fresh-statistics plan-shape match, Panel B reports recovery of stale/fresh mismatches, and Panel C reports new deviations where stale already matched fresh.

Finally, we check that adding the consistency-gated soft projection to the calibrated router introduces no plan-shape regression here. On a six-configuration planner subset (left/right/bimodal shift $\times$ two seeds, 504 queries), the calibrated router is byte-identical to the residual-gated Hybrid—row Q-error 2.234, fresh-plan match 90.1%, new plan deviations 8.17%—because the gate routes to ISOMER on every configuration and never selects soft. For reference, hard projection alone is 2.237/90.9%/6.86% and soft projection alone is 2.277/91.1%/6.54% on the same subset. The residual gate’s preference for ISOMER over hard projection on these planner-facing cases is a property of the existing Hybrid, not of the soft candidate: adding soft to the pool is plan-shape-safe by construction, and the router’s single-column and optimizer-facing gains (Section 5.4) come at no planner cost relative to the deployed Hybrid.

6.7. Feedback Budget and Noise Sensitivity

The deployment form of OASIS depends on two practical conditions: how many recent feedback observations are available and how accurate those feedback observations are. To stress these conditions without running additional DBMS loads, we use the optimizer-decision proxy described in Section 4. All methods see the same feedback window and future predicates; only the feedback budget or feedback values are changed.

Panels A–B of Figure 6 vary the number of visible feedback observations.

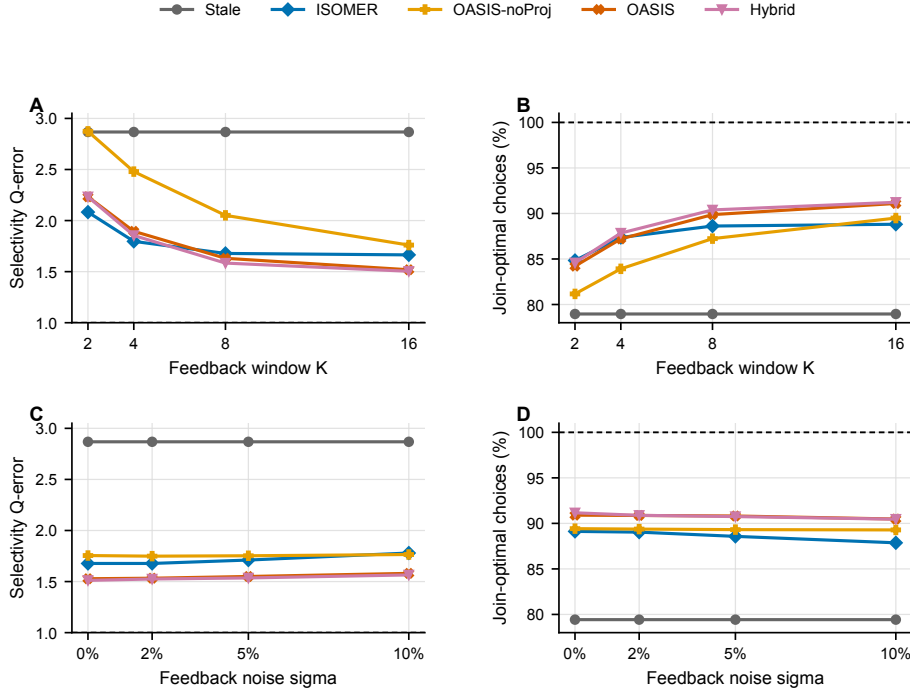


Figure 6: Feedback-budget and feedback-noise sensitivity in the optimizer-decision proxy. Panels A–B vary the visible feedback window K ; Panels C–D perturb feedback selectivities with multiplicative log-normal noise while holding future predicates fixed. Dashed lines denote fresh statistics.

With only two observations, consistency projection is helpful but limited: the full OASIS reaches selectivity Q-error 2.233 and Hybrid matches 84.6% of optimal join choices, compared with stale Q-error 2.867 and 79.0% join-optimal choices. As the window grows, the learned correction and projection become more complementary. At $K = 16$, the full OASIS reaches Q-error 1.518 and Hybrid reaches 1.504, with 91.1%–91.2% join-optimal choices. Hybrid’s selections also show that no single component dominates all cases: at $K = 16$, it selects ISOMER for 37% of cases, the full OASIS for 47%, OASIS-noProj for 14%, and stale for 2%.

Panels C–D perturb feedback selectivities with multiplicative log-normal noise while keeping the future predicates and true selectivities fixed. The diagnostic uses 64 held-out cases per drift intensity and one noise seed. At 10% feedback noise, the full OASIS still improves selectivity Q-error from

Table 13: Deployment safety checks. Projection and residual-based Hybrid gating are evaluated without oracle/fresh information. P/I/O/S denote OASIS (full two-stage), ISOMER, OASIS-noProj, and stale choices in the residual gate.

Check	Risk signal	Unguarded state	Calibrated/gated state	Gate signal
PostgreSQL planner	New plan deviations	OASIS-noProj 13.8%	OASIS 1.1%; FreshPlan 96.1%	projection
PostgreSQL joins	Join new deviations	OASIS-noProj 19.8%	OASIS 5.0%; Recovery 90.7%	projection
Sparse feedback ($K = 2$)	Join-optimal choices	Stale 79.0%	Hybrid 84.6%; NewRisk 2.5%	I 27%, P 57%, O 14%, S 1%
Full feedback ($K = 16$)	Join-optimal choices	Stale 79.0%	Hybrid 91.2%; NewRisk 1.4%	I 37%, P 47%, O 14%, S 2%
10% feedback noise	Join-optimal choices	Stale 79.4%	Hybrid 90.5%; NewRisk 1.7%	I 36%, P 42%, O 18%, S 3%
DML trace sanity	Selectivity Q-error	Stale 1.295	Hybrid 1.072; OASIS 1.076	I 77%, P 20%, O 2%, S 1%

2.868 to 1.580 and maintains 90.5% join-optimal choices; Hybrid reaches Q-error 1.565 with the same join-optimal rate. ISOMER is more sensitive to noise in this setting, rising from Q-error 1.677 at clean feedback to 1.779 at 10% noise. The result supports the role of the projection stage as a calibration layer, but it also motivates residual-based gating and abstention when feedback attribution is unreliable.

6.8. Deployment Safety Summary

Table 13 collects the safety evidence that determines how OASIS should be deployed. The learned stage alone (OASIS-noProj) improves row-estimation accuracy, but it can introduce plan deviations when the stale planner already agreed with fresh statistics. The projection stage reduces these new deviations from 13.8% to 1.1% in the PostgreSQL batch and from 19.8% to 5.0% on join queries. In the optimizer-decision proxy, residual-gated Hybrid improves join-optimal choices under sparse feedback, full feedback, and 10% feedback noise while keeping new risky decisions near 1–3%. On the benchmark-inspired DML traces, Hybrid improves aggregate selectivity Q-error from 1.295 to 1.072 and selects ISOMER or the full OASIS in 97% of cases. These results make the projection stage and residual gating part of the method, rather than optional post-processing.

7. Overhead and Deployment Considerations

Per-correction OASIS inference averages 1.06 ms in our implementation, including tensorization and one forward pass. The PostgreSQL-style statistics-format conversion adds 0.066–0.073 ms, for a combined correction cost of approximately 1.13 ms. Feedback collection adds bounded bookkeeping per filter conjunct and emits one observation tuple per conjunct at query completion. These measurements support the intended use case: maintaining

optimizer-facing statistics between scheduled statistics refreshes, not replacing full refreshes after major schema changes or bulk data shifts.

8. Related Work

Classical histogram construction builds statistics from data scans [5, 6], while streaming summaries such as KLL sketches provide compact approximate quantiles [7]. Adaptive selectivity estimation has long used query feedback to refine estimation functions [8]. Feedback-driven histograms, including self-tuning histograms, STHoles, and ISOMER, refine statistics from observed query results [9, 1, 2]. OASIS differs by learning a reusable drift-correction function that emits DBMS-consumable statistics rather than applying only local per-table refinement rules or returning point selectivity estimates.

Learned cardinality estimators such as NeuroCard, Naru, DeepDB, FACE, and MSCN predict query cardinalities using learned density or representation models [10, 11, 12, 13, 14]. These systems are complementary to OASIS: they often output point estimates for specific query classes, while OASIS outputs corrected marginal statistics that can be consumed by an existing CBO. Copula-based methods and factorized join estimators such as FactorJoin [4] similarly combine one-dimensional marginals with a dependence or join-key model; our composition-family and FactorJoin experiments embed OASIS as the marginal source in exactly these estimators and show that marginal quality and feedback consistency are prerequisites for such downstream use.

Plan-level optimization systems such as LEO, Neo, and Bao adjust plan choices or optimizer behavior based on execution history [15, 16, 17]. OASIS operates below that layer by correcting the statistics object itself. A system could combine both approaches: statistics correction supplies better base-table inputs, while plan-level learning handles residual cost-model or join-order effects.

9. Limitations

The evaluation is limited to tested synthetic drift families, benchmark-inspired DML traces, and controlled PostgreSQL planner-only scenarios. The DML traces make insert/update/delete event streams explicit, but they are not production logs or DBMS telemetry; validating the same correction policy on production statistics histories and workload traces remains future work.

The PostgreSQL batch covers two drift directions, two table sizes, and three random seeds, so it validates the statistics-to-planner path more robustly than a single case, but it does not establish wall-clock runtime improvement. Runtime depends on executor behavior, caching, physical design, system load, and cost-model calibration. OASIS also does not infer multi-column dependence from single-column feedback. When downstream estimates depend on correlations, an external dependence model or multivariate statistics source is still required. Finally, feedback itself can be noisy because of predicate attribution, sampling, or concurrent updates. Our noise diagnostic shows graceful degradation up to 10% multiplicative feedback noise, but deployments should pair correction with residual-based gating, abstention, and rollback policies. The calibrated router also gates on feedback residual, which is a proxy for estimate quality rather than for plan-shape risk: on the PostgreSQL subset it prefers ISOMER and so inherits the Hybrid’s slightly higher new-deviation rate (8.17% versus 6.86% for hard projection alone), even though it never routes to the soft candidate there. A plan-shape-aware gating criterion that could route to hard projection over ISOMER on planner-facing cases is left to future work.

10. Conclusion

OASIS repairs stale single-column optimizer statistics from query feedback without rescanning tables or modifying the optimizer. The main lesson is that feedback-driven statistics repair should be separated into two questions: how to generate a useful marginal prior, and how to calibrate that imperfect prior so it is safe for optimizer consumption. The learned OASIS MLP answers the first question for our system by extrapolating drift beyond the observed feedback window. Under a fixed projection operator, the learned prior improves held-out future-predicate Q-error over projection from stale statistics, and it is especially useful on strong out-of-distribution batch drift.

The paper’s deployment contribution is the second question: regime-aware feedback calibration. A raw repaired marginal, even a useful learned one, need not be locally consistent with the predicates just observed. Downstream estimators amplify that inconsistency: OASIS-noProj gives weak and unstable multi-column gains and is actively harmful in the bilinear FactorJoin join kernel. Hard feedback projection makes the marginal safe for composition and planner-facing use, improving every member of a six-estimator composition family by 22%–38%, recovering stable FactorJoin gains, and re-

Table A.14: Feedback-locality diagnostic on the mixed predicate population. Held-out future predicates are binned by the directed Hausdorff distance from their endpoints to the nearest feedback endpoint in fresh-CDF space. Values are geometric-mean selectivity Q-error; ISOMER is projection from a stale start and OASIS is the full two-stage system. OASIS win is the per-predicate fraction on which the full OASIS beats ISOMER.

Locality	Preds	Stale	ISOMER	OASIS-noProj	OASIS	Fresh	OASIS vs ISOMER	OASIS win
near $\leq$ 0.05	33012	2.159	1.441	1.487	1.352	1.000	+6.1%	58.5%
mid $\leq$ 0.15	21712	2.546	1.574	1.777	1.495	1.000	+5.0%	53.6%
far $>$ 0.15	10812	3.054	1.779	1.918	1.660	1.000	+6.7%	52.2%

ducing PostgreSQL new plan deviations from 13.8% to 1.1% while lowering aggregate row-estimation Q-error from 27.748 to 2.360.

Calibration itself is regime-dependent. Hard projection is the right anchor for joins and the DBMS planner, whereas a consistency-gated soft projection generalizes better on future single-column and optimizer-facing predicates. The residual-gated calibrated router resolves this split using only deployment-visible feedback residuals: it lowers single-column Q-error from 1.442 to 1.395 and optimizer-proxy selectivity Q-error from 1.446 to 1.396 over hard projection, while reducing to hard projection or ISOMER on the join kernel and PostgreSQL planner where soft projection is not selected. The banded projection negative result shows that the improvement is not a consequence of loosening constraints; it comes from choosing the right calibration regime. Finally, the Stage-1 prior-source swap shows that this layer is not tied to the OASIS MLP: the same Stage-2 router improves stale, heuristic, STHoles, QuickSel-H, and learned priors in aggregate. This makes OASIS a practical feedback-calibrated statistics layer for maintaining optimizer-facing marginals between scheduled `ANALYZE` passes. No query runtime improvement is measured or claimed.

Appendix A. Feedback-Locality Diagnostic

To see where the learned stage’s advantage comes from, we bin the held-out future predicates of Section 6.2 by their distance to the feedback window. Each predicate endpoint is mapped to fresh-CDF space and assigned the directed Hausdorff distance to the nearest feedback endpoint, giving near (≤ 0.05), mid (≤ 0.15), and far (> 0.15) bins.

Table A.14 reports the diagnostic on the mixed predicate population. The full OASIS beats ISOMER in every bin, and—restricting to the operative

drift regime $q \geq 5$ —by +9.2%, +6.9%, and +8.0% in the near, mid, and far bins. The advantage is therefore not confined to predicates near the feedback window. The raw learned marginal (OASIS-noProj) is, by contrast, worse than ISOMER in every bin and worst in the far bin, confirming that raw learned marginals are not reliably deployable across locality bins; the projection turns the learned initialization into a stable statistics object.

The far bin is predicate-type dependent. For a threshold-only population (pure $</>$ predicates, which probe only a cumulative CDF level), the far-bin advantage shrinks to +0.8% at $q \geq 5$, while near and mid remain strong (+7.7% and +5.4%). Range and point predicates, which probe local density rather than a cumulative level, retain the far-bin advantage. The learned stage’s value in uncovered regions therefore comes from better local density, which helps range and point queries but adds little to purely cumulative threshold queries.

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